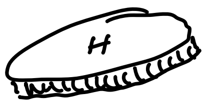
Binomial Distribution

$$p(y|\theta) = \binom{n}{y} \theta^{y} (1-\theta)^{n-y}$$

$$\binom{n}{y} = \frac{n!}{y!(n-y)!}$$



$\theta = 0.75$ $\quad H = 1$

$\quad\quad\quad\quad\; T = 0$

What is the probability of TTT?

$$p(y=0, n=3 \mid \theta = 0.75) = \binom{3}{0} (0.75)^0 (0.25)^3 \approx 1.6\%$$

$$\rightarrow \quad \underline{p(y=1, n=3} \mid \theta = 0.75) = \binom{3}{1} (0.75)^1 (0.25)^2 \approx 14\%$$

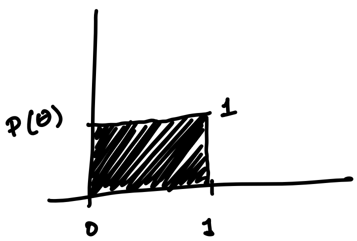
$\rightarrow \quad \{ TTH, \; THT, \; HTT \}$

Binomial Likelihood:

$$p(y|\theta) = \binom{n}{y}\theta^{y}(1-\theta)^{n-y}$$

Prior:

$$p(\theta) = 1 \qquad 0 \le \theta \le 1$$



Posterior:

$$p(\theta|y) \propto p(\theta)\, p(y|\theta)$$

constant w.r.t. θ

$$= \cancel{\binom{n}{y}}\theta^{y}(1-\theta)^{n-y} \qquad 0 \le \theta \le 1$$

$$p(\theta|y) \propto \underline{\theta^{y}(1-\theta)^{n-y}}$$

Once normalized

$$p(\theta|y) = B(\underbrace{y+1}_{\alpha}, \underbrace{n-y+1}_{\beta})$$

$$p(\tilde{y} \mid y) = \int p(\tilde{y}, \theta \mid y)\, d\theta$$

$$= \int p(\tilde{y} \mid \theta, y)\, p(\theta \mid y)\, d\theta$$

chain rule of joint probabilities

$$= \int \underbrace{p(\tilde{y} \mid \theta)}_{\text{prediction}}\ \underbrace{p(\theta \mid y)}_{\text{posterior}}\, d\theta$$

$\rightarrow$ posterior predictive distribution

$$p(\tilde{y} \mid \theta) = \binom{m}{\tilde{y}} \theta^{\tilde{y}} (1-\theta)^{m-\tilde{y}}$$

$$p(\theta \mid y) = \text{Beta}(y+1, n-y+1) = \frac{\Gamma(n+2)}{\Gamma(y+1)\,\Gamma(n-y+1)} \theta^{y} (1-\theta)^{n-y}$$

$$p(\tilde{y} \mid y) = \int \frac{\Gamma(n+2)}{\Gamma(y+1)\,\Gamma(n-y+1)} \cdot \binom{m}{\tilde{y}} \cdot \theta^{y+\tilde{y}} (1-\theta)^{m+n-(\tilde{y}+y)}\, d\theta$$

$$= \binom{m}{\tilde{y}} \frac{\Gamma(n+2)}{\Gamma(y+1)\,\Gamma(n-y+1)} \cdot \frac{\Gamma(y+\tilde{y}+1)\,\Gamma(m+n-\tilde{y}-y+1)}{\Gamma(m+n+2)}$$

$$\times \underbrace{\cancel{\int \frac{\Gamma(m+n+2)}{\Gamma(y+\tilde{y}+1)\,\Gamma(m+n-\tilde{y}-y+1)} \theta^{y+\tilde{y}} (1-\theta)^{m+n-(\tilde{y}+y)}\, d\theta}}_{\text{Beta distribution}} \rightarrow 1$$

$$= \binom{m}{\tilde{y}} \frac{\Gamma(n+2)}{\Gamma(y+1)\,\Gamma(n-y+1)} \frac{\Gamma(y+\tilde{y}+1)\,\Gamma(m+n-\tilde{y}-y+1)}{\Gamma(m+n+2)} \cdot 1$$

$$= \binom{m}{\tilde{y}} \frac{(n+1)!\,(y+\tilde{y})!\,(m+n-\tilde{y}-y)!}{y!\,(n-y)!\,(m+n+1)!}$$

$$= \binom{m}{\tilde{y}} \frac{\overset{(n+1)\,n!}{(n+1)!}\;(y+\tilde{y})!\;(m+n-\tilde{y}-y)!}{y!\;(n-y)!\;\underset{(m+n+1)(m+n)!}{(m+n+1)!}}$$

$$= \binom{m}{\tilde{y}} \frac{(n+1)}{m+n+1} \cdot \frac{n!}{y!(n-y)!} \cdot \frac{(y+\tilde{y})!\,(m+n-(y+\tilde{y}))!}{(m+n)!}$$

$$= \binom{m}{\tilde{y}} \binom{n}{y} \left[\binom{m+n}{y+\tilde{y}}\right]^{-1} \cdot \frac{n+1}{m+n+1}$$

n = # of data points

y = # of successes

m = # of predicted points

$\tilde{y}$ = # of predicted successes

Prediction: next birth is female $\quad m = \tilde{y} = 1$

$$p(\tilde{y}=1, m=1 \mid y) = \binom{n}{y} \left[\binom{n+1}{y+1}\right]^{-1} \cdot \frac{n+1}{n+2}$$

$$= \frac{\cancel{n!}}{\cancel{y!(n-y)!}} \cdot \frac{\overset{y+1}{\cancel{(y+1)!}}\,\cancel{(n-y)!}}{\cancel{(n+1)!}} \cdot \frac{\cancel{n+1}}{n+2} = \frac{\overset{\text{data}}{y}\overset{\text{mean of posterior}}{+1}}{n+2}$$

$$= \frac{y+1}{n+2}$$

$\text{Beta}(\alpha, \beta)$ have a mean of $\dfrac{\alpha}{\alpha+\beta}$

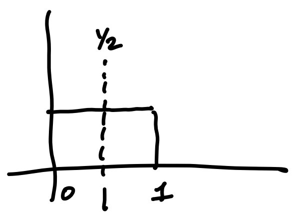
What is the mean of the posterior distribution?

$$p(\theta|y) = \text{Beta}(y+1, n-y+1)$$

$E(\theta|y)$ is the mean

$$= \frac{y+1}{n-y+1+y+1} = \frac{y+1}{n+2}$$

if n is close to 0, then the average is more influenced by the average of the posterior distribution



As n gets large,

$$E(\theta|y) \sim \frac{y}{n}$$